

RecallRoom

A reviewable response to an ingredient recall

Product brief / Shivam Gupta / recallroom.web.app

The problem and the buyer

A supplier names one ingredient lot. A small food manufacturer must work out which batches and customer shipments it reached. The relevant evidence may sit in receiving sheets, production records and shipment exports. RecallRoom gives the quality manager a private case to reconcile those records and expose what still needs review.

The workflow

Upload records and retain the originals. NVIDIA Nemotron 3 Super on Nebius Token Factory uses a schema-guided prompt to extract cited proposals. Application checks validate drafts and calculate lot scope deterministically. A reviewer approves records, investigates ambiguity and exports the packet. The real hosted workflow passed reviewed import, reload and inspected exports.

The model interprets records. It cannot approve a recall decision or silently release a hold. Customer messages remain drafts that the application never sends.

A fictional drill with a consequential ambiguity

Lot PB-0901-A	Synthetic result
Confirmed finished products	480 oat bars + 320 energy bites = 800 units
Confirmed disposition	600 shipped to 3 customers; 200 on site
Separate cookie uncertainty	240 units, including 120 shipped
Lot A ingredient remaining	40-50 kg before review; 50 kg after correction

The cookie shorthand could name either received peanut-butter lot. A reviewed supporting record assigns it to lot B. The decision log preserves that change. Ingredient kilograms and finished-unit counts remain separate. These are demonstration facts, not customer outcomes or a general safety clearance.

The commercial hypothesis

Start with recurring traceability drills for small manufacturers and co-packers. Test **\$99 per site per month**, with food-safety consultants as a possible distribution channel. The price, recurring use and willingness to pay remain hypotheses. No customer interviews, paid pilots or measured savings have been completed.

FoodDocs and Mar-Kov already provide traceability capabilities. RecallRoom proposes a narrower incident workspace over existing exports and documents. Potential defensibility would come from reliable document adapters, reviewed lot aliases and a difficult evaluation set. Onboarding and support costs may determine viability more than inference cost.

Why this workflow deserves testing

FDA's 2026 tabletop exercise involved 15 voluntary companies. It found gaps in information retention and sharing, while many relevant facts existed in traditional business records. The sample does not represent the whole industry. RecallRoom targets evidence quality and readiness; it does not claim automatic FDA compliance. [1]

Implemented MVP and current proof boundary

Firebase Hosting serves the app. Firebase Authentication supports Google and email/password sign-in. A Cloud Run API verifies identity and controls Firestore investigations and private Google Cloud Storage originals. Cloud Tasks runs authenticated extraction jobs. PDF text requires human verification. Reviewed decisions and HTML, CSV or JSON packets preserve the evidence.

41 automated tests, 60 hosted API checks and 10 operator checks pass. These cover application behavior using synthetic records and mocked provider contracts. **The hosted NVIDIA workflow passed.** Inference: 17.562 seconds, 2,600 input and 4,203 output tokens, about \$0.00456 estimated cost. Earlier omissions prompted coverage checks. One fictional scenario does not establish real-world accuracy. Scanned-image OCR is outside the MVP.

The next validation

Evaluate more source files against human answer keys, including ambiguity and duplicate imports. Observe quality managers completing a drill. Measure review effort and unsupported exclusions. Test paid pilots and repeat usage.

Primary sources and project

[1] FDA Traceability Readiness Tabletop Exercises Final Report (2026)

[2] FoodDocs: product and pricing

[3] Mar-Kov: manufacturing traceability

Public demo: recallroom.web.app

Public repository and live evaluation trace

Created by Shivam Gupta, with AI-assisted research, implementation, testing and documentation.